

CHEMICAL ANALYSIS – CHROMATOGRAPHY

KEY VOCABULARY

Stationary phase: the chromatography paper (does not move).

Mobile phase: the solvent that moves through the paper.

Rf value: distance by spot $\div$ distance by solvent.

Pure substance: produces a single spot on a chromatogram.

COMMON MISCONCEPTIONS

~~Use a pen to draw the start line.~~

✓ Use pencil — ink would separate too.

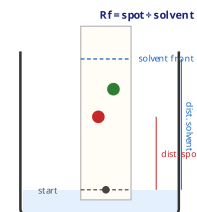
~~Rf can be greater than 1.~~

✓ Rf is always between 0 and 1.

~~The solvent level starts above the spots.~~

✓ Solvent must start BELOW the spots.

PAPER CHROMATOGRAPHY



1 TWO PHASES

Name the **stationary** and **mobile** phases in paper chromatography. [2 marks]

2 WHY PENCIL

Explain why the start line is drawn in pencil. [1 mark]

3 CALCULATE Rf

A spot moves 4.0 cm; the solvent moves 8.0 cm. Calculate the Rf value. [2 marks]

Answer: _____ (Rf)

4 PURE OR IMPURE

Explain how a chromatogram shows a substance is a mixture. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

A more ____ substance travels further. $Rf = \text{distance by } ___ \div \text{distance by solvent}$.

soluble

spot

insoluble

solvent

6 INTERPRET

Explain why some substances travel further up the paper. [2 marks]
